Problem Set 7

Due Saturday, October 31, 11:59 PM.

You may use Mathematica or other computational tools in all problem sets.

1. (a) The Hamiltonian operator for a particle of mass m constrained to undergo circular motion on a frictionless track is given by $\hat{H} = -\frac{\hbar^2}{2mR^2}\frac{\partial^2}{\partial\phi^2}$, where R is the radius of the circle and ϕ is the angular coordinate. What boundary condition should the wavefunction $\psi(\phi)$ satisfy? Using the Schrödinger equation and the boundary condition, determine the wavefunctions and energies of the stationary states. Determine the degeneracy of each energy level. (5 points)

   (b) Consider the three-dimensional harmonic oscillator with potential $V(x, y, z) = \frac{1}{2}m\omega^2(x^2 + y^2 + z^2)$. Answer the following questions using cartesian coordinates, separation of variables, and your knowledge of the energies and wavefunctions of the one-dimensional simple harmonic oscillator. Write down the energy levels in terms of three quantum numbers n_1, n_2, and n_3. Write the same energy levels in terms of one quantum number n. Show that the n^{th} energy level is $\frac{1}{2}(n+1)(n+2)$−fold degenerate. (5 points)

2. Consider a "rigid rotor" model for the rotational motion of H_2 molecule with two H atoms, each of mass M, at fixed separation of $2R$.

   (a) Considering a molecule rotating about an axis perpendicular to the center of the interatomic separation, write down the classical Hamiltonian for the rotational energy of the rotor in terms of angular momentum $\vec{L}$ and moment of inertia $I = 2MR^2$. (5 points)

   (b) Promote the classical Hamiltonian to a Hamiltonian operator, and write down the wavefunctions in spherical coordinates and the corresponding energies. (5 points)

   (c) Using the mass of the proton for the mass of the hydrogen atom and taking $2R = 0.15\,nm$, determine the numerical value of the first four energy levels in eV. (5 points)

3. (a) Show that the radial part of the Schödinger equation for the electron of mass m in a hydrogen atom with potential $V = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r}$; i.e., $-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left(-\frac{e^2}{4\pi\epsilon_0}\frac{1}{r} + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right)u = Eu$, where $u(r) = rR(r)$, can be written as $\frac{d^2u}{d\rho^2} = \left(1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2}\right)u$, where $\rho = \kappa r$, $\rho_0 = \frac{me^2}{2\pi\epsilon_0\hbar^2\kappa}$, $\kappa = \sqrt{-\frac{2mE}{\hbar^2}}$, and l is the total angular quantum number. (5 points)

   (b) Investigate the asymptotic large and small ρ behaviors of the radial equation and show that normalizable solutions could be written as $u(\rho) = \rho^{l+1}e^{-\rho}v(\rho)$ with the equation $\rho\frac{d^2v}{d\rho^2} + 2(l + 1 - \rho)\frac{dv}{d\rho} + (\rho_0 - 2(l+1))v = 0$ for $v(\rho)$. (5 points)

(c) Show that the method of power series with $v(\rho) = \sum_{j=0}^{\infty} c_j \rho^j$ produces the recursion relations $c_{j+1} = \frac{2(j+l+1)-\rho_0}{(j+1)(j+2l+2)} c_j$. (5 points)

(d) Using the large j behavior of the recursion relation, argue how the infinite series would give $v(\rho) \sim e^{2\rho}$ and reintroduce a divergent solution. (5 points)

(e) Truncating the series, show that the energy levels are quantized and given by $E_n = -\frac{me^4}{32\pi^2\epsilon_0^2\hbar^2 n^2} = -\frac{13.6\,eV}{n^2}$, where $n = 1, 2, 3, \cdots$. Show that the n^{th} energy level is $2n^2-$fold degenerate. (5 points)

(f) Make a plot of the four lowest energy levels to scale, and indicate the list of associated quantum numbers (nlm_l) for each energy level. (5 points)

(g) Assuming that a hydrogen atom making a transition from a higher energy level with principal quantum number m to a lower energy level n gives off all its energy by releasing a photon, write down the expression for the frequency of the emitted radiation. Show that the possible frequencies can be written as $f = R_\infty\, c\,(\frac{1}{n^2} - \frac{1}{m^2})$, where $R_\infty \approx 1.1 \times 10^7 m^{-1}$ is the Rydberg constant and c is the speed of light. (5 points)

4. How long did it take you to complete this problem set?